

Hospital Management System (HMS) - High-Level Summary

The **Hospital Management System (HMS)** is a software application designed to simplify and automate hospital administrative tasks, specifically **patient registration, appointment management, and medication tracking**. The system aims to improve the efficiency, accuracy, and accessibility of hospital operations while minimizing manual paperwork.

Overall Concept

The HMS allows hospital staff (administrators, receptionists, and doctors) to manage patient information and appointments in a centralized system. Patients can be registered with unique identifiers, scheduled for appointments with available doctors, and have their medications prescribed and recorded. The system ensures data consistency, reduces scheduling conflicts, and provides a clear record of patient history.

High-Level Operation

- 1. Patient Registration:** A new patient's personal and medical details are entered into the system. Each patient is assigned a unique ID for easy identification. Present patient information can be updated or retrieved for reference.
- 2. Appointment Booking:** Receptionists or patients can book appointments with doctors. The system checks doctor availability to avoid scheduling conflicts. Appointments can be rescheduled or canceled if needed. Notifications (optional) can inform staff of upcoming appointments.
- 3. Medication Management:** After consultation, doctors can prescribe medications for patients. Prescriptions are logged in the system along with dosage and instructions. Patient's medication history is maintained for future reference.

Explanation of the Flow

Patient Registration -> Appointment Booking -> Consultation -> Medication Prescription

Patient Registration: A new patient enters personal and contact details. The system assigns a **unique patient ID**.

Patient Info Stored in DB: All details are saved securely in the hospital database.

Appointment Booking: Staff or the patient selects a doctor and preferred time.

Check Doctor Availability: The system verifies the chosen time slot.

Appointment Confirmed: The booking is stored, and confirmation may be sent to staff/patient.

Consultation: The patient sees the doctor at the scheduled time.

Prescription/Medication: Doctor prescribes medication, which is logged in the patient's record.

Medication History Stored: Future consultations can access the patient's medication history.